STARlab Capital — ML Internship Task

1. Document Details

- **Date:** 9 March 2026
- **Owner:** STARlab Capital
- **Intern:** Mohd. Abuzar

2. Purpose

The purpose of this project is to design and implement a **Market Anomaly Detection system** using statistical and machine learning techniques on stock market data.

The system will identify anomalous trading periods that deviate from normal market behavior. The project will use **AAPL stock data as the primary case study**.

Dataset access is available at the provided link. (Link to Dataset)

3. Scope

The intern will:

- Explore and understand the dataset structure and characteristics
- Perform exploratory data analysis (EDA) to identify patterns and irregularities
- Engineer **time-aware features**
- Implement statistical and machine learning–based anomaly detection methods
- Perform time-aware validation of detected anomalies
- Track experiments and document findings

Out of scope:
Production deployment and real-time integration.

4. Dataset & Target Definition

The intern must document:

- Data source and description
- Time range covered by the dataset
- Data frequency
- Key variables such as:
    - price (open, high, low, close)
    - bid and ask quotes
    - trade volume (not to detect anomalies for this feature)
    - VWAP
- Definition of **anomaly signals or abnormal behavior**

5. Data Preprocessing

Define and implement preprocessing steps including:

- Handling missing or inconsistent values
- Performing data quality checks (invalid prices, bid–ask inconsistencies)
- Time alignment and resampling where necessary.
- Initial outlier detection and cleaning

All preprocessing steps must be **fully reproducible**.

6. Feature Engineering

- Create features using only **historical data**.
- All engineered features must avoid **future data leakage**.

7. Evaluation Strategy

The evaluation will focus on:

- inspection of high anomaly scores
- visualization of anomalies in price/volume time series
- identification of abnormal market events

8. Experiment Tracking

Maintain a structured experiment log including:

- Detection method used
- Feature set
- Anomaly score distribution
- Number and type of detected anomalies
- Key observations

9. Error Analysis

Perform detailed analysis of detected anomalies:

- investigate extreme anomaly scores
- analyze time windows with abnormal activity
- determine whether anomalies correspond to:
  - genuine market events
  - noise or data errors
  - model limitations

10. Final Deliverables

Submit:

- Clean, modular, and well-documented code
- Data preprocessing and feature engineering pipeline
- Implemented anomaly detection models
- Experiment tracking logs
- Visualization of detected anomalies
- Final report summarizing findings and insights

11. Recommended Reading

Forecasting: Principles and Practice
https://otexts.com/fpp2/

Selected Videos from:

Time Series Analysis
